ZS-S3

Modified Gravity Phenomenology: Horndeski Embedding, G_eff Framework, and Observational Pull Summary

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Verification: 25/25 PASS | Zero Free Parameters

§0. Abstract

We classify the Z-Spin action within the Horndeski scalar-tensor framework: $G_4 = M^2_P(1+A|\Phi|^2)/2$, $G_3 = G_5 = 0$, ensuring gravitational-wave speed $c_T = c$ (GW170817-compatible, structural). The effective gravitational parameters are $\mu = 1/(1+A) = 0.926$, $\eta = 1$ (no slip), $\Sigma = 1/(1+A) = 0.926$. The radial-mode mass $m_\rho \sim O(M_P)$ (ZS-F1 v1.0 §4.4) yields Compton wavelength $\lambda_C \approx 3.9\times10^{-34}$ m, suppressing any fifth force to undetectable levels ($r/\lambda_C \gtrsim 10^{38}$ for $r \gtrsim 10$ km). The G_eff cancellation in the growth equation [PROVEN in ZS-F3 v1.0] ensures all S_8 suppression arises from Ω_m^eff shift alone. An observational pull summary covering 10 predictions shows 9 of 10 within 1.6σ (zero free parameters), with D/H resolved to −0.05σ (ZS-U4 v1.0 §6). Verification: 25/25 PASS.

Keywords: Horndeski gravity, modified gravity parameters, gravitational-wave speed, fifth force, effective gravitational constant, scalar-tensor phenomenology

Epistemic Status Legend

Status	Definition
PROVEN	Follows from standard mathematics alone. Machine-verifiable.
DERIVED	Follows from Z-Spin action + standard physics. Zero free parameters.
VERIFIED	Numerically confirmed against observational data or independent computation.
TESTABLE	Well-defined prediction awaiting experimental data.
STANDARD	Accepted SM/cosmological input (not Z-Spin specific).
NON-CLAIM	Explicitly not asserted. Documented to prevent overclaim.
RETRACTED	Previously claimed, now withdrawn with documented reason.

§1. Introduction

The Z-Spin action $S = \int d^4x \sqrt{(-g)}\, [(M^2_P/2)(1 + A\varepsilon^2)R - (M^2_P/2)(\nabla\varepsilon)^2 - V(\varepsilon)]$ defines a scalar-tensor theory with non-minimal coupling strength $A = 35/437$. Any scalar-tensor theory must be classified within the Horndeski framework to establish (i) gravitational-wave speed compatibility with GW170817,

(ii) effective gravitational parameters for cosmological surveys, and (iii) fifth-force suppression for laboratory and solar-system tests.

This paper performs that classification and presents the resulting modified-gravity phenomenology. ZS-S3 is the CANONICAL source for the Horndeski embedding, the effective gravitational parameters (μ, η, Σ), the fifth-force analysis, and the observational pull summary. For the foundational derivations of cosmological predictions (H_0 = e^A, S_8 mechanism, $w_0 = -1$, Ω_m^eff), see ZS-F3 v1.0. For the global fit with all 10 observables and falsification protocol, see ZS-U4 v1.0.

§2. Locked Inputs (from ZS-F3 v1.0 and ZS-F2 v1.0)

All observational predictions are derived in ZS-F3 v1.0. This paper imports the results:

Prediction	Value	Equation	Source
A	35/437	ZS-F2 v1.0	**LOCKED**
H_0^{loc}/H_0^{cMB}	e^A = 1.0834	ZS-F3 v1.0 Eq.(6–7)	**DERIVED**
H_0^{pred}	72.98 ± 0.59 km/s/Mpc	ZS-F3 v1.0 Eq.(8)	**DERIVED**
Ω_m^eff	0.2984	ZS-F3 v1.0 Eq.(11)	**DERIVED**
S_8	0.794 (4.5% suppression)	ZS-F3 v1.0 §4	**DERIVED**
w_0	−1 (exact)	ZS-F3 v1.0 Eq.(13)	**DERIVED**
G_eff	G/(1+A) = 0.9258G	ZS-F3 v1.0 Eq.(4–5)	**DERIVED**
m_ρ	0.1602 M_P	ZS-F1 v1.0 §4.4	**DERIVED**

No re-derivation appears in ZS-S3. For full derivations, see ZS-F3 v1.0.

§3. Horndeski Embedding (Canonical)

The Z-Spin action maps to the Horndeski scalar-tensor framework:

$$G_2 = -(M^2_P/2)X - V(\Phi), \quad G_3 = 0 \quad (1)$$

$$G_4 = (M^2_P/2)(1 + A|\Phi|^2), \quad G_5 = 0 \quad (2)$$

Key consequences:

(a) $G_5 = 0 \Rightarrow$ c_T = c exactly (no GW speed anomaly). Structural, not tuned.

(b) $G_3 = 0 \Rightarrow$ no kinetic braiding. Shift-symmetric in θ-sector (Goldstone).

(c) $G_4 \neq M^2_P/2 \Rightarrow$ modified gravity strength G_eff = G/(1+A).

[STATUS: DERIVED] *Horndeski classification from Z-Spin action. $G_5 = 0$ ensures GW170817 compatibility structurally.*

3.1 GW170817 Constraint: c_T = c

The gravitational-wave speed constraint $|c_T/c - 1| < 3\times10^{-15}$ (LIGO/Virgo) eliminates all Horndeski theories with $G_5 \neq 0$. Z-Spin satisfies this structurally: $G_5 = 0$ is not a fine-tuning but a direct consequence of the minimal coupling $(1+A\varepsilon^2)R$ having no derivative coupling to curvature.

[STATUS: PROVEN] *No derivative coupling ⇒ $G_5 = 0$ ⇒ c_T = c. Structural.*

§4. Effective Gravitational Parameters

In the quasistatic subhorizon limit, the Z-Spin modified gravity is characterized by three parameters:

$\mu \equiv G_eff/G = 1/(1+A) = 0.9258$ *[Poisson equation strength]* (3)

$\eta \equiv \Psi/\Phi = 1$ *[gravitational slip, no anisotropic stress]* (4)

$\Sigma \equiv G_light/G = 1/(1+A) = 0.9258$ *[lensing strength]* (5)

Physical interpretation: Z-Spin is a conformal scalar-tensor theory with universal coupling. Both Poisson and lensing equations are modified identically ($\mu = \Sigma$), producing zero gravitational slip ($\eta = 1$). This is the simplest non-trivial modified gravity: Brans-Dicke with $\omega_BD = \infty$ (frozen scalar at attractor).

[STATUS: DERIVED] *From Horndeski mapping. η = 1 ensures no ISW/lensing discrepancy.*

4.1 G_eff Cancellation in Growth Equation

The linear growth equation $Source/H^2 = 4\pi G_eff\rho_m / H^2 = (3/2)\Omega_m(a)$ shows that G_eff cancels between the Poisson equation numerator and the Friedmann equation denominator. This is proven algebraically in ZS-F3 v1.0 §4.3. Consequence: ALL S_8 suppression arises from $\Omega_m\text{^}eff = 0.2984 < 0.3153$, not from modified gravity per se.

This cancellation explains why Z-Spin predicts 4.5% S_8 suppression (background shift) rather than the invalid 8.7% scale-dependent mechanism retracted from earlier internal notes.

§5. Fifth Force Analysis

The radial-mode perturbation $\delta\varepsilon$ around the attractor $\varepsilon = 1$ has mass:

$m_\rho = \sqrt{(2\lambda_vac)}\, M_P \sim O(M_P)$ (6)

where $\lambda_vac \sim O(1)$ is the vacuum-scale self-coupling (ZS-F1 v1.0 §4.4). From ZS-U5 v1.0 §8: $\lambda_vac = 2A^2 = 0.01283$, giving $m_\rho = 2A \times M_P = 0.1602\ M_P$. Fifth-force suppression holds for any $\lambda_vac > 10^{-120}$.

Compton wavelength: $\lambda_C = \hbar/(m_\rho\, c) = \hbar/(2A\, M_P\, c) \approx 3.9\times10^{-34}$ m.

For any macroscopic distance r ≳ 10 km:

$$r/\lambda_C \gtrsim 10^{38} \Rightarrow \text{Yukawa suppression} \sim e^{-r/\lambda_C} \approx 0 \quad (7)$$

The fifth force is suppressed by a factor of $e^{-10^{38}}$ — observationally zero to any conceivable precision. No Vainshtein screening or chameleon mechanism is needed; the Planck-scale mass provides natural suppression.

[STATUS: DERIVED] *From ZS-U1 v1.0 radial-mode analysis. No screening mechanism required.*

Solar-system and laboratory tests:

Test	Z-Spin prediction	Bound	Status
PPN $\gamma-1$	~0 (Yukawa killed)	$< 2.3\times10^{-5}$ (Cassini)	PASS
PPN $\beta-1$	~0	$< 8\times10^{-5}$ (perihelion)	PASS
$\lvert c_T/c - 1\rvert$	0 (structural)	$< 3\times10^{-15}$ (GW170817)	PASS
Lunar laser ranging	~0	$< 10^{-13}$	PASS

§6. Observational Pull Summary (10 Predictions)

Complete pull analysis against current data. For derivations see ZS-F3 v1.0; for BBN details and global fit see ZS-U4 v1.0.

Observable	Z-Spin	Observed	Error	Pull (σ)	Source
H_0 ratio	1.0834	1.0832±0.012	SH0ES	−0.06	ZS-F3 v1.0
Ω_m^{eff}	0.2984	0.2975±0.0086	DESI DR2	+0.11	ZS-F3 v1.0
w_0	−1.000	−1.055±0.036	DESI	+1.53	ZS-F3 v1.0
S_8 (DES Y3)	0.794	0.776±0.017	DES	+1.06	ZS-F3 v1.0
S_8 (KiDS)	0.794	0.766±0.020	KiDS-1000	+1.40	ZS-F3 v1.0
$\Omega_b h^2$	0.02250	0.02237±0.00015	Planck	+0.87	ZS-F3 v1.0
n_s	0.9674	0.9649±0.0042	Planck	+0.60	ZS-U1 v1.0
r	0.0089	<0.036	BK18	PASS	ZS-U1 v1.0
Y_p	0.2410	0.2449±0.004	Aver+ 2015	−0.98	ZS-U4 v1.0
D/H $\times10^5$	2.526	2.527±0.030	Cooke+ 2018	−0.05	ZS-U4 v1.0 §6

Summary: 9/10 within 1.6σ. D/H resolved to −0.05σ (ZS-U4 v1.0 §6, Z-corrected).

Zero free parameters beyond A = 35/437. The H_0 ratio prediction exp(A) is exact (δA = 0). The absolute value H_0^{pred} = 72.98 ± 0.59 km/s/Mpc inherits σ from H_0^{cMB} propagation (ZS-F3 v1.0 Eq.8).

§7. Consolidation Note: F3 ↔ S3 ↔ U4 Role Matrix

Topic	ZS-F3 v1.0 (Derivation)	ZS-S3 v1.0 (Phenomenology)	ZS-U4 v1.0 (Global Fit)
H_0	CANONICAL derivation	Value in pull table	Scoreboard entry
S_8	CANONICAL mechanism	G_eff cancellation ref	Survey comparison
w_0	CANONICAL proof	Value in pull table	DESI context
Horndeski	—	CANONICAL embedding	—
μ, η, Σ	—	CANONICAL parameters	—
Fifth force	—	CANONICAL analysis	—
BBN	—	Pull values only	CANONICAL analysis
Falsification	Appendix (5 gates)	MG-specific gates	CANONICAL protocol

§8. Falsification Conditions (MG-Specific)

Gate	Condition	Observable	Status
FS3-1	$\lvert c_T/c - 1\rvert > 10^{-15}$ detected	GW events	**PROVEN** (= 0)
FS3-2	$\eta \neq 1$ (gravitational slip detected at >3σ)	Euclid/LSST	**TESTABLE**
FS3-3	Fifth force detected at r > 1 mm	Lab experiments	**TESTABLE**
FS3-4	Pull table: >2 observables outside 3σ	Combined	**TESTABLE**
FS3-5	Horndeski classification changes ($G_5 \neq 0$ required)	Theoretical	**PROVEN** (= 0)

§9. Conclusion

The Z-Spin action is classified within the Horndeski scalar-tensor framework as a $G_4 = (M^2_P/2)(1+A|\Phi|^2)$ theory with $G_3 = G_5 = 0$, structurally ensuring c_T = c and eliminating the need for fine-tuning against GW170817. The effective gravitational parameters $\mu = \Sigma = 1/(1+A) = 0.9258$, $\eta = 1$ represent the simplest non-trivial modified gravity consistent with all current data. The Planck-scale radial-mode mass $m_\rho = 0.1602\ M_P$ provides natural fifth-force suppression ($\lambda_C \approx 3.9\times10^{-34}$ m) without screening mechanisms. The G_eff cancellation in the growth equation ensures S_8 suppression arises solely from background Ω_m^eff shift. An observational pull summary of 10 predictions shows 9/10 within 1.6σ with zero free parameters.

§10. Verification Suite (25/25 PASS)

Category	Tests	Pass/Fail	Key Result
H_0 Mapping & Dark Energy	5	5/0	H_0=72.98, w_0=−1, DESI compatible
Cosmic Density Budget	5	5/0	Ω_b=6/121, Ω_m^eff=0.2984
S_8 Suppression & Growth	5	5/0	S_8=0.794, G_eff cancellation, surveys
G_eff Framework & Horndeski	5	5/0	μ=0.926, η=1, c_T=c, fifth force
BBN, Inflation & Falsification	5	5/0	Y_p, n_s, r, D/H, 6 gates
TOTAL	**25**	**25/0**	**100% pass rate**

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

Verification script: ZS_S3_verify_v1_0.py. Dependencies: Python 3.10+, NumPy. Execution: python3 ZS_S3_verify_v1_0.py. Expected output: 25/25 PASS, exit code 0. The verification suite is publicly available.

Appendix A: Cross-Reference Table

Paper	Content	Direction	Relation
ZS-F3 v1.0	H_0, S_8, w_0, Ω_m derivations	ZS-F3 → ZS-S3 §2	UPSTREAM
ZS-F2 v1.0	A = 35/437	Input → ZS-S3	LOCKED
ZS-F1 v1.0 §4.4	m_ρ ~ O(M_P) (λ_vac)	Input → ZS-S3 §5	DERIVED
ZS-U1 v1.0	Inflation: n_s, r	Input → ZS-S3 §6	DERIVED
ZS-U4 v1.0	Global fit, BBN, falsification	ZS-S3 → ZS-U4 (exports)	DOWNSTREAM
ZS-U5 v1.0	λ_vac = $2A^2$	Input → ZS-S3 §5	DERIVED
ZS-A3 v1.0	GW predictions	ZS-S3 → ZS-A3 (c_T)	DOWNSTREAM

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0.)

Horndeski embedding: $G_4 = M^2_P(1+A|\Phi|^2)/2$, $G_3 = G_5 = 0$ ($c_T = c$ structural). Effective parameters: $\mu = \Sigma = 1/(1+A) = 0.9258$, $\eta = 1$. Fifth force: $m_\rho = 0.1602\ M_P$, $\lambda_C \approx 3.9\times10^{-34}$ m (natural suppression). Observational pull summary: 10 predictions, 9/10 within 1.6σ, D/H resolved to −0.05σ (ZS-U4 v1.0 §6). G_{eff} cancellation proven. Solar-system tests: all PASS. F3↔S3↔U4 role matrix. 5 MG-specific falsification gates. Verification: 25/25 PASS. Zero free parameters.

Z-Sim cross-reference (March 2026): All 8 closure parameters DERIVED from A = 35/437 and (Z,X,Y) = (2,3,6). See ZS-Q7 v1.0, ZS-M3 v1.0, ZS-T3 v1.0.